G6_Task

Hello,

Please read the following carefully:

As a Software Engineering research team at the Polytechnique Montréal (Canada), we are interested in understanding developers' assessment of coding task difficulty. You will be provided with a series of 12 coding questions. Questions are divided into two parts:

- First, you will be given three instructions representing coding tasks and you will be asked to write, in a few words, what is the underlying task represented by all these instructions. The instructions represent the same tasks, but they vary in wording and information they give. In a nutshell, you have to distill the essence of the task and write it in a few words.

- In a second time, you will be asked to assess how difficult implementing this ***task*** would be. Note we ask the assessment based on the task you will have worded in the previous part, not based on any of the instructions presented earlier. First, you will be asked your subjective judgment on a scale from Very Easy to Very Difficult. Then, to make judging the difficulty more relatable, we adopt a similar approach as what is used in agile development and planning poker, i.e. you will be asked to evaluate how long it would take to implement a task (in minutes here). Consider in your assessment the knowledge needed to implement the task, the external information you would need to use, whether you would need the assistance of a LLM such as ChatGPT (and so the time it would take to debug the obtained code) etc. Finally, you will have to explain in a few words why you gave such difficulty ratings.

To give you an idea of the range of tasks you will need to assess, the first section of this questionnaire will be two examples we devised to illustrate the questionnaire.

Completing the task will take about 30 minutes.

Note also that all examples are written using Python 3.10 syntax.

Please make sure to answer all the questions to the best of your ability.

Please note that your identity and personal data will not be disclosed, while we plan to use the aggregated results and anonymized responses as part of a scientific publication. If you have any questions about the questionnaire or our research, please do not hesitate to contact us.

By proceeding, you consent to take part in the study under the conditions described previously.

Thank you,

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* Indique une question obligatoire

1. **Please enter your prolific ID :** *

Example of tasks

As we ask of you some estimation of difficulty and the tasks dealt in this questionnaire can be a bit different from what you can be used to as a developer, we give you examples of some tasks along with some assessments.

Note that this assessment is **our assessment**. You may have a different opinion based on your background. The idea of those examples is to give an extreme range of what type of tasks you can expect in the questionnaire.

As a reminder: You will first be given three instructions and asked to explain what is the task is about in a few words. Then, you will have to rate the difficulty of the ***task*** you just described (not accounting for particular instruction). Rating the difficulty is done in two steps: subjective difficulty and approximate time needed. Finally, you will be asked to say why you gave such rating.

Example 1

Instruction 1

```python
def subtract(c1, c2):
        """
        Deliver the result of subtracting two complex numbers,
        specified as 'c1' and 'c2', as a complex output.
        :param c1: The first complex number,complex.
        :param c2: The second complex number,complex.
        :return: The difference of the two complex numbers,complex.
        """
```

Instruction 2

```python
def subtract(c1, c2):
        """
        The method receives two complex numbers 'c1' and 'c2', from which it
        subtracts 'c2.real' from 'c1.real' to determine the real portion, and
        subtracts 'c2.imag' from 'c1.imag' to determine the imaginary portion.
        It constructs and returns a new complex number using these results
        with the 'complex()' function.
        :param c1: The first complex number,complex.
        :param c2: The second complex number,complex.
        :return: The difference of the two complex numbers,complex.
        """
```

Instruction 3

```python
def subtract(c1, c2):
        """
        Subtracts two complex numbers.
        :param c1: The first complex number,complex.
        :param c2: The second complex number,complex.
        :return: The difference of the two complex numbers,complex.
        >>> complexCalculator = ComplexCalculator()
        >>> complexCalculator.subtract(1+2j, 3+4j)
        (-2-2j)
        """
```

Answer

The task encapsulated by the **subtract**

function

is to compute the subtraction between two complex numbers.

This task has a subjective difficulty of **Very Easy** as assessed by us and would take about **1 min**
to implement. There is no particular difficulty, the task is very
straightforward without any corner case, it just requires to know what a
complex is, and the standard Python library for it.

Example 2

Context

```python
import numpy as np
class MetricsCalculator2:
    """
    The class provides to calculate Mean Reciprocal Rank (MRR) and Mean Average Precision (MAP)
    based on input data, where MRR measures the ranking quality and MAP measures the average precision.
    """

    def __init__(self):
        pass

    @staticmethod
    def map(data):
        pass
```

Instruction 1

```python
def mrr(data):
        """
        Compute the MRR of the input data. MRR is a widely used evaluation index.
        It is the mean of reciprocal rank.
        :param data: the data must be a tuple, list 0,1,eg.([1,0,...],5).
        In each tuple (actual result,ground truth num),ground truth num is the total ground num.
        ([1,0,...],5), or list of tuple eg. [([1,0,1,...],5),([1,0,...],6),([0,0,...],5)].
        1 stands for a correct answer, 0 stands for a wrong answer.
        :return: if input data is list, return the recall of this list. if the input data is list of list, return the
        average recall on all list. The second return value is a list of precision for each input.
        >>> MetricsCalculator2.mrr(([1, 0, 1, 0], 4))
        >>> MetricsCalculator2.mrr([([1, 0, 1, 0], 4), ([0, 1, 0, 1], 4)])
        1.0, [1.0]
        0.75, [1.0, 0.5]

        """
```

Instruction 2

```python
def mrr(data):
        """
        Assess the Mean Reciprocal Rank (MRR) from the input 'data'.
        MRR assesses the average of reciprocal ranks of outcomes.
        The input structure 'data' must be a tuple consisting of a list of
        binary values and its total count or a list of such tuples.
        Each binary value (0 or 1) represents the correctness of an answer.
        If 'data' is a tuple, the function returns its mean reciprocal rank.
        If 'data' is a list, the function returns the overall average MRR along
        with a list showing reciprocal ranks for each component tuple.

        :param data: the data must be a tuple, list 0,1,eg.([1,0,...],5).
        In each tuple (actual result,ground truth num),ground truth num is the total ground num.
        ([1,0,...],5), or list of tuple eg. [([1,0,1,...],5),([1,0,...],6),([0,0,...],5)].
        1 stands for a correct answer, 0 stands for a wrong answer.
        :return: if input data is list, return the recall of this list. if the input data is list of list, return the
        average recall on all list. The second return value is a list of precision for each input.

        """
```

Instruction 3

```
def mrr(data):
        """
        Evaluate the Mean Reciprocal Rank (MRR) from 'data'. This metric calculates the mean of reciprocal
        ranks for correct answers. Input 'data' could be either a tuple containing binary correctness indicators
        and the aggregate count of answers, or multiple tuples of this type in a list. For each tuple, the function
        measures reciprocal ranks by multiplying the binary indicators by reciprocals of their positions, finding the
        earliest nonzero value to establish the rank of the first correct response. If the input 'data' is a tuple,
        the MRR calculated for this specific tuple is returned; if it's a list, the function summarizes by providing
        both the average MRR and a list of individual MRR values for each tuple.

        :param data: the data must be a tuple, list 0,1,eg.([1,0,...],5).
        In each tuple (actual result,ground truth num),ground truth num is the total ground num.
        ([1,0,...],5), or list of tuple eg. [([1,0,1,...],5),([1,0,...],6),([0,0,...],5)].
        1 stands for a correct answer, 0 stands for a wrong answer.
        :return: if input data is list, return the recall of this list. if the input data is list of list, return the
        average recall on all list. The second return value is a list of precision for each input.

        """
```

Answer

The task encapsulated by the **mrr** function is to compute the
Mean Reciprocal Rank of binary tuples data.

This task has a subjective difficulty of **Very Hard** as assessed by us and would take about
40 min
to implement. The difficulty mainly branches from dealing with the
tuple data and how the formula is processed accordingly (with a few corner cases). This can take
additional time if the Mean Reciprocal Rank is not known.

Question 1:

Below are instructions for implementing the function.

Instruction 1

```
def decimal_to_binary(decimal):
    """
    Construct a function named 'decimal_to_binary' which is designed to accept
    an integer in decimal form. This function needs to convert the decimal to a
    binary representation, stripping away the '0b' prefix typically seen in Python.
    Wrap this resultant binary string between 'db' at both its opening and closing,
    and return the customized string format.
    """
```

Instruction 2

```python
def decimal_to_binary(decimal):
    """
    You will be given a number in decimal form and your task is to
    convert it to binary format. The function should return a string,
    with each character representing a binary
    number. Each character in the string will be '0' or '1'.

    There will be an extra couple of characters 'db' at the beginning
    and at the end of the string.
    The extra characters are there to help with the format.

    Examples:
    decimal_to_binary(15)   # returns "db1111db"
    decimal_to_binary(32)   # returns "db100000db"
    """
```

Instruction 3

```python
def decimal_to_binary(decimal):
    """
    Define a function called 'decimal_to_binary' which receives a parameter 'decimal'.
    Utilize the 'bin()' function in Python to transform 'decimal' into a binary string.
    Ensure to remove the prefix '0b' from the binary output, then add 'db' to both the
    beginning and ending of the string before returning it.
    """
```

2. In a few words, what is the task this function aims to implement? *

3. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- Very Easy
- Easy
- Not Easy Nor Difficult
- Difficult
- Very Difficult

4. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- 1 min
- 5 min
- 10 min
- 20 min
- 40 min
- > 40 min

5. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

<u>Question 2:</u>

Below are instructions for implementing the function:

Context

```python
class CamelCaseMap:
    """
    This is a custom class that allows keys to be in camel case style by converting them
    from underscore style, which provides dictionary-like functionality.
    """

    def __init__(self):
        """
        Initialize data to an empty dictionary
        """
        self._data = {}

    def __getitem__(self, key):
        pass

    def __delitem__(self, key):
        pass

    def __iter__(self):
        pass

    def __len__(self):
        pass

    def _convert_key(self, key):
        pass

    @staticmethod
    def _to_camel_case(key):
        pass
```

Instruction 1

```python
def __setitem__(self, key, value):
        """
        Update the '_data' dictionary by inserting the 'value' at a modified 'key',
        ensuring first to apply camel case conversion to the 'key' using '_convert_key'.
        This procedure maintains consistency in key formatting throughout the dictionary,
        adapting keys entered in underscore style to camel case in accordance with the
        class’s settings.
        :param key:str
        :param value:str, the specified value
        :return:None
        """
```

Instruction 2

```python
def __setitem__(self, key, value):
        """
        Set the value corresponding to the key to the specified value
        :param key:str
        :param value:str, the specified value
        :return:None
        >>> camelize_map = CamelCaseMap()
        >>> camelize_map['first_name'] = 'John'
        >>> camelize_map.__setitem__('first_name', 'new name')
        camelize_map['first_name'] = 'new name'
        """
```

Instruction 3

```
def __setitem__(self, key, value):
        """
        Alter the dictionary '_data' by assigning the given 'value' to the
        corresponding 'key', which is first transformed into camel case with
        the '_convert_key' method. It guarantees that keys presented in
        different styles are unified in camel case format for storing, aligning
        with the class's design of uniform key formatting.
        :param key:str
        :param value:str, the specified value
        :return:None
        """
```

6. In a few words, what is the task this function aims to implement? *

7. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- Very Easy
- Easy
- Not Easy Nor Difficult
- Difficult
- Very Difficult

8. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- 1 min
- 5 min
- 10 min
- 20 min
- 40 min
- > 40 min

9. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

<u>Question 3:</u>

Below are instructions for implementing the function:

Instruction 1

```python
def strange_sort_list(lst):
    """
    Design a function named 'strange_sort_list' that receives a list of
    integers 'lst'. The function will extract and return integers in
    a unique pattern: first by selecting the smallest integer, then the
    largest from the remaining integers, continuing alternately until
    all are included. The integers are sorted in an ascending list named
    'sorted_list'. Pointers 'i' (start) and 'j' (end) track the positions
    within 'sorted_list', populating the list 'ans' alternately from both
    ends using a while loop until 'i' is less than 'j'. If 'i' equals 'j'
    at the end of the loop, the final central integer is appended to 'ans'.
    'ans' is then returned, featuring the integers in this specific sequence.
    """
```

Instruction 2

```python
def strange_sort_list(lst):
    """
    Given list of integers, return list in strange order.
    Strange sorting, is when you start with the minimum value,
    then maximum of the remaining integers, then minimum and so on.

    Examples:
    strange_sort_list([1, 2, 3, 4]) == [1, 4, 2, 3]
    strange_sort_list([5, 5, 5, 5]) == [5, 5, 5, 5]
    strange_sort_list([]) == []
    """
```

Instruction 3

```python
def strange_sort_list(lst):
    """
    Develop the function 'strange_sort_list' which receives a list of integers
    as an argument. This function is designed to create a new list where numbers
    are arranged alternately, beginning with the smallest, then the largest of
    the remaining, and continues in this manner. Initially, the input list is
    sorted in increasing order, and with two pointers starting at the ends of
    this sorted list, numbers are chosen alternately and compiled into the
    resultant list until it contains all elements.
    """
```

10. In a few words, what is the task this function aims to implement? *

11. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- Very Easy
- Easy
- Not Easy Nor Difficult
- Difficult
- Very Difficult

12. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- 1 min
- 5 min
- 10 min
- 20 min
- 40 min
- > 40 min

13. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 4:

Below are instructions for implementing the function:

Context

```python
class BankAccount:
    """
    This is a class as a bank account system, which supports deposit money,
    withdraw money, view balance, and transfer money.
    """

    def __init__(self, balance=0):
        """
        Initializes a bank account object with an attribute balance, default value is 0.
        """
        self.balance = balance

    def deposit(self, amount):
        pass

    def view_balance(self):
        pass

    def transfer(self, other_account, amount):
        pass
```

Instruction 1

```python
def withdraw(self, amount):
    """
    Withdraws a certain amount from the account, decreasing
    the account balance, return the current account balance.
    If amount is negative, raise a ValueError("Invalid amount").
    If the withdrawal amount is greater than the account balance,
    raise a ValueError("Insufficient balance.").
    :param amount: int
    """
```

Instruction 2

```python
def withdraw(self, amount):
    """
    Initiate by validating whether the withdrawal 'amount' is
    less than zero. If this is the case, prompt a 'ValueError'
    with a 'Invalid amount' remark. Proceed to ascertain if the
    'amount' overreaches 'self.balance', whereupon another
    'ValueError' should be raised bearing 'Insufficient balance.'.
    Passing these assessments, reduce 'self.balance' by the stated
    'amount', and furnish the revised balance.
    :param amount: int"
    """
```

Instruction 3

```python
def withdraw(self, amount):
    """
    Start by evaluating if the withdrawal 'amount' is negative and if confirmed,
    raise a 'ValueError' labeled 'Invalid amount'. Then, check if the 'amount'
    is greater than 'self.balance', and if true, raise another 'ValueError'
    titled 'Insufficient balance.'. Successfully passing these checks, the
    'amount' should be subtracted from 'self.balance', thereafter returning
    the updated account balance.
    :param amount: int
    """
```

14. In a few words, what is the task this function aims to implement? *

____

15. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ⬭ Very Easy
- ⬭ Easy
- ⬭ Not Easy Nor Difficult
- ⬭ Difficult
- ⬭ Very Difficult

16. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- 1 min
- 5 min
- 10 min
- 20 min
- 40 min
- > 40 min

17. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

<u>Question 5:</u>

Below are instructions for implementing the function:

Instruction 1

```python
def largest_prime_factor(n: int):
    """
    Design a function 'largest_prime_factor' to receive an integer 'n'
    greater than 1, ensuring 'n' isn't prime. It calculates the highest
    prime factor of 'n'. Initially, it marks primes up to 'n' using a
    boolean list and then checks backwards from 'n-1' to identify the
    greatest prime factor that divides 'n' exactly, and returns it.
    """
```

Instruction 2

```python
def largest_prime_factor(n: int):
    """
    Return the largest prime factor of n. Assume n > 1
    and is not a prime.
    >>> largest_prime_factor(13195)
    29
    >>> largest_prime_factor(2048)
    2
    """
```

Instruction 3

```python
def largest_prime_factor(n: int):
    """
    Develop a function called 'largest_prime_factor' that identifies
    and returns the highest prime factor of an integer 'n', where 'n'
    is greater than 1 and not a prime.
    """
```

18. In a few words, what is the task this function aims to implement? *

19. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- Very Easy
- Easy
- Not Easy Nor Difficult
- Difficult
- Very Difficult

20. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- 1 min
- 5 min
- 10 min
- 20 min
- 40 min
- > 40 min

21. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 6:

Below are instructions for implementing the function:

Context

```python
class Interpolation:
    """
    This is a class that implements the Linear interpolation
    operation of one-dimensional and two-dimensional data
    """

    def __init__(self):
        pass
    @staticmethod
    def interpolate_2d(x, y, z, x_interp, y_interp):
        pass



```

Instruction 1

```python
def interpolate_1d(x, y, x_interp):
        """
        Perform linear interpolation of one-dimensional data using
        the function "interpolate_1d", which accepts three parameters:
        "x", "y", and "x_interp". The lists "x" and "y" represent the
        original data points' x and y coordinates, while "x_interp"
        contains the x-coordinates at which the values need to
        be interpolated. The function initializes an empty list
        "y_interp". For each x-coordinate "xi" in "x_interp",
        the function loops through the "x" list to find the segment
        where "xi" fits between "x[i]" and "x[i+1]". It then applies
        the linear interpolation formula:
        "yi = y[i] + (y[i+1] - y[i]) * (xi - x[i]) / (x[i+1] - x[i])"
        to compute the interpolated y-value "yi", which is then appended
        to "y_interp". Finally, the list "y_interp", containing all
        interpolated y-values, is returned.
        :param x: The x-coordinate of the data point, list.
        :param y: The y-coordinate of the data point, list.
        :param x_interp: The x-coordinate of the interpolation point, list.
        :return: The y-coordinate of the interpolation point, list.
        """
```

Instruction 2

```python
def interpolate_1d(x, y, x_interp):
        """
        Linear interpolation of one-dimensional data
        :param x: The x-coordinate of the data point, list.
        :param y: The y-coordinate of the data point, list.
        :param x_interp: The x-coordinate of the interpolation point, list.
        :return: The y-coordinate of the interpolation point, list.
        >>> interpolation = Interpolation()
        >>> interpolation.interpolate_1d([1, 2, 3], [1, 2, 3], [1.5, 2.5])
        [1.5, 2.5]
        """
```

Instruction 3

```
def interpolate_1d(x, y, x_interp):
        """
        Develop a function for linear interpolation based on 1D data using two
        coordinate lists 'x' and 'y' for the data points. An additional list,
        'x_interp', is used to indicate positions where interpolation should occur.
        The function should return a list, 'y_interp', which will consist of the
        y-values interpolated at the positions specified by 'x_interp'.
        :param x: The x-coordinate of the data point, list.
        :param y: The y-coordinate of the data point, list.
        :param x_interp: The x-coordinate of the interpolation point, list.
        :return: The y-coordinate of the interpolation point, list.
        """
```

22. In a few words, what is the task this function aims to implement? *

____

23. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- Very Easy
- Easy
- Not Easy Nor Difficult
- Difficult
- Very Difficult

24. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- 1 min
- 5 min
- 10 min
- 20 min
- 40 min
- > 40 min

25. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

<u>Question 7:</u>

Below are instructions for implementing the function:

Instruction 1

```python
def fix_spaces(text):
    """
    Design a function called 'fix_spaces' that converts a
    supplied string by changing all single spaces to underscores.
    Moreover, any sequence of spaces longer than two in the
    input string should be substituted with a dash. The function
    carries out this operation by initially searching for the
    maximum length of consecutive spaces, progressing downwards to
    three, and replacing such groups with a dash before switching
    any leftover spaces to underscores.
    """
```

Instruction 2

```python
def fix_spaces(text):
    """
    Given a string text, replace all spaces in it with underscores,
    and if a string has more than 2 consecutive spaces,
    then replace all consecutive spaces with -

    fix_spaces("Example") == "Example"
    fix_spaces("Example 1") == "Example_1"
    fix_spaces(" Example 2") == "_Example_2"
    fix_spaces(" Example   3") == "_Example-3"
    """
```

Instruction 3

```python
def fix_spaces(text):
    """
    Define a function 'fix_spaces' that adjusts a string by
    converting every space to an underscore. Replace any
    series of more than two consecutive spaces with a single dash.
    """
```

26. In a few words, what is the task this function aims to implement? *

27. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- Very Easy
- Easy
- Not Easy Nor Difficult
- Difficult
- Very Difficult

28. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- 1 min
- 5 min
- 10 min
- 20 min
- 40 min
- > 40 min

29. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 8:

Below are instructions for implementing the function:

Context

```python
import openpyxl
class ExcelProcessor:
    """
    This is a class for processing excel files, including readring and writing excel data,
    as well as processing specific operations and saving as a new excel file.
    """

    def __init__(self):
        pass

    def read_excel(self, file_name):
        pass

    def process_excel_data(self, N, save_file_name):
        pass



```

Instruction 1

```python
def write_excel(self, data, file_name):
        """
        Develop a function that writes to an Excel file by taking 'data' and
        'file_name' as parameters, where 'data' consists of a list of tuples
        (each tuple is a row) and 'file_name' is a string denoting the Excel
        file's name. Start by creating a workbook and activating the initial
        sheet. Loop through each tuple in 'data', appending them to the sheet.
        Save the workbook using 'file_name', close it, and return 1 on
        successful operation or 0 if an error occurs.
        :param data: list, Data to be written
        :param file_name: str, Excel file name to write to
        :return: 0 or 1, 1 represents successful writing, 0 represents failed writing
        """
```

Instruction 2

```
def write_excel(self, data, file_name):
        """
        Write data to the specified Excel file
        :param data: list, Data to be written
        :param file_name: str, Excel file name to write to
        :return: 0 or 1, 1 represents successful writing,
        0 represents failed writing
        >>> processor = ExcelProcessor()
        >>> new_data = [
        >>> ('Name', 'Age', 'Country'),
        >>> ('John', 25, 'USA'),
        >>> ('Alice', 30, 'Canada'),
        >>> ('Bob', 35, 'Australia'),
        >>> ('Julia', 28, 'Germany')
        >>> ]
        >>> data = processor.write_excel(new_data, 'test_data.xlsx')
        """
```

Instruction 3

```
def write_excel(self, data, file_name):
        """
        Create a function that inputs data into a specified Excel document,
        taking 'data' and 'file_name' as parameters. Here, 'data' should be
        a list of tuples, each tuple acting as a row, and 'file_name' a
        string indicating the Excel file's intended name. The function
        kick-starts by creating a new workbook with 'openpyxl.Workbook()'
        and accessing the active sheet immediately with 'workbook.active'.
        As it cycles through each tuple in 'data', it appends each to the
        sheet using 'sheet.append(row)'. After writing all data, it secures
        the workbook with 'workbook.save(file_name)' and closes it using
        'workbook.close()'. It concludes either with a return value of 1
        if successful or 0 if it encounters any exceptions.
        :param data: list, Data to be written
        :param file_name: str, Excel file name to write to
        :return: 0 or 1, 1 represents successful writing, 0 represents failed writing
        """
```

30. In a few words, what is the task this function aims to implement? *

31. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- Very Easy
- Easy
- Not Easy Nor Difficult
- Difficult
- Very Difficult

32. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- 1 min
- 5 min
- 10 min
- 20 min
- 40 min
- > 40 min

33. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

<u>Question 9:</u>

Below are instructions for implementing the function:

Instruction 1

```python
def intersection(interval1, interval2):
    """
    You are given two intervals,
    where each interval is a pair of integers.
    For example, interval = (start, end) = (1, 2).
    The given intervals are closed which means that the interval (start, end)
    includes both start and end.
    For each given interval, it is assumed that its start is less or equal its end.
    Your task is to determine whether the length of intersection of these two
    intervals is a prime number.
    Example, the intersection of the intervals (1, 3), (2, 4) is (2, 3)
    which its length is 1, which not a prime number.
    If the length of the intersection is a prime number, return "YES",
    otherwise, return "NO".
    If the two intervals don't intersect, return "NO".


    [input/output] samples:
    intersection((1, 2), (2, 3)) ==> "NO"
    intersection((-1, 1), (0, 4)) ==> "NO"
    intersection((-3, -1), (-5, 5)) ==> "YES"
    """
```

Instruction 2

```python
def intersection(interval1, interval2):
    """
    Create a function called 'intersection' accepting two tuples, 'interval1' and 'interval2',
    representing two intervals with integer start and end points. Initially, the function checks
    and swaps the intervals to make sure 'interval1' starts before 'interval2'. It then determines
    if the intervals overlap and calculates the overlap length by subtracting the start of
    'interval2' from the lesser endpoint of the two intervals. An inner function 'is_prime(a)'
    assesses whether this length is prime, by asserting 'a' is greater than one and has no divisors
    from 2 to its square root. The main function will return 'YES' if the length is prime, otherwise
    'NO'. If there's no overlap, it returns 'NO'.
    """
```

Instruction 3

```python
def intersection(interval1, interval2):
    """
    Write a function named "intersection" that determines if the
    length of the intersection of two given intervals is a prime number.
    Each interval is a pair of integers, representing the start and end
    points. If the length of this intersection is prime, return "YES",
    otherwise return "NO". If there is no intersection, also return "NO".
    """
```

34. In a few words, what is the task this function aims to implement? *

35. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- Very Easy
- Easy
- Not Easy Nor Difficult
- Difficult
- Very Difficult

36. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- 1 min
- 5 min
- 10 min
- 20 min
- 40 min
- > 40 min

37. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 10:

Below are instructions for implementing the function:

Context

```
import numpy as np
class KappaCalculator:
    """
    This is a class as KappaCalculator, supporting to calculate Cohen's
    and Fleiss' kappa coefficient.
    """

    @staticmethod
    def fleiss_kappa(testData, N, k, n):
        pass
```

Instruction 1

```
def kappa(testData, k):
        """
        Calculate Cohen's kappa coefficient from a 'testData' matrix with k
        dimensions for assessing inter-rater accord. Convert 'testData' into
        a numpy structure 'dataMat', and set 'P0' to zero. While iterating
        over k, sum the diagonal to update 'P0', indicating exact rater
        agreements. Compute sums of matrix rows ('xsum') and columns ('ysum'),
        and an overall sum ('sum'). Expected chance agreement 'Pe' is calculated
        thereby by multiplying 'xsum' and 'ysum' and dividing by 'sum ** 2'.
        'P0' is divided by the 'sum'. The kappa formula ('P0' - 'Pe') / (1 - 'Pe')
        is used to obtain and return the final coefficient as a float.
        :param testData: The k-dimensional matrix that needs to calculate the
                                                       cohens kappa value
        :param k: int, Matrix dimension
        :return:float, the cohens kappa value of the matrix
        """
```

Instruction 2

```
def kappa(testData, k):
    """
    Calculate the cohens kappa value of a k-dimensional matrix
    :param testData: The k-dimensional matrix that needs to calculate
                                              the cohens kappa value
    :param k: int, Matrix dimension
    :return:float, the cohens kappa value of the matrix
    >>> KappaCalculator.kappa([[2, 1, 1], [1, 2, 1], [1, 1, 2]], 3)
    0.25
    """
```

Instruction 3

```
def kappa(testData, k):
    """
    From a k-dimensional matrix 'testData', compute and return
    the Cohen's kappa coefficient.
    :param testData: The k-dimensional matrix that needs to
    calculate the cohens kappa value
    :param k: int, Matrix dimension
    :return:float, the cohens kappa value of the matrix
    """
```

38. In a few words, what is the task this function aims to implement? *

39. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- Very Easy
- Easy
- Not Easy Nor Difficult
- Difficult
- Very Difficult

40. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- 1 min
- 5 min
- 10 min
- 20 min
- 40 min
- > 40 min

41. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 11:

Below are instructions for implementing the function

Instruction 1

```python
from typing import List, Tuple

def find_closest_elements(numbers: List[float]) -> Tuple[float, float]:
    """
    Create a function 'find_closest_elements' which accepts a list of numbers.
    The function's purpose is to identify and return in ascending order the two
    numbers in the list that have the least difference between them. Initially,
    the function sorts the list of numbers. Subsequently, it loops through the
    sorted list, computing the gap between successive numbers and records the
    smallest gap encountered along with the corresponding numbers pair.
    Ultimately, the pair with the least difference is returned.
    """
```

Instruction 2

```python
from typing import List, Tuple

def find_closest_elements(numbers: List[float]) -> Tuple[float, float]:
    """
    From a supplied list of numbers (of length at least two) select
    and return two that are the closest to each
    other and return them in order (smaller number, larger number).
    >>> find_closest_elements([1.0, 2.0, 3.0, 4.0, 5.0, 2.2])
    (2.0, 2.2)
    >>> find_closest_elements([1.0, 2.0, 3.0, 4.0, 5.0, 2.0])
    (2.0, 2.0)
    """
```

Instruction 3

```python
from typing import List, Tuple

def find_closest_elements(numbers: List[float]) -> Tuple[float, float]:
    """
    Write a function named 'find_closest_elements' that identifies and returns
    a pair of numbers from a list that are closest to each other.
    The function should return the pair in ascending order.
    """
```

42. In a few words, what is the task this function aims to implement? *

43. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- Very Easy
- Easy
- Not Easy Nor Difficult
- Difficult
- Very Difficult

44. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- 1 min
- 5 min
- 10 min
- 20 min
- 40 min
- > 40 min

45. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 12:

Below are instructions for implementing the function:

Context

```python
class GomokuGame:
    """
    This class is an implementation of a Gomoku game, supporting for making moves,
    checking for a winner, and checking if there are five consecutive symbols on
    the game board.
    """

    def __init__(self, board_size):
        """
        Initializes the game with a given board size.
        It initializes the board with empty spaces and sets the current player symble as 'X'.
        """
        self.board_size = board_size
        self.board = [[' ' for _ in range(board_size)] for _ in range(board_size)]
        self.current_player = 'X'

    def make_move(self, row, col):
        pass

    def check_winner(self):
        pass



```

Instruction 1

```python
def _check_five_in_a_row(self, row, col, direction):
        """
        checks if there are five consecutive symbols of the same player in a row starting from a
        given cell in a given direction (horizontal, vertical, diagonal).
        Counts the number of consecutive symbols in that direction starting from the given cell,
        :param row: int, row of the given cell
        :param col: int, column of the given cell
        :param direction: tuple, (int, int), named as (dx, dy). Row and col will plus several dx
        and dy repectively.
        :return: True if there are five consecutive symbols of the same player, and False otherwise.
        >>> gomokuGame = GomokuGame(10)
        >>> moves = [(5, 5), (0, 0), (5, 4), (0, 1), (5, 3), (0, 2), (5, 2), (0, 3), (5, 1)]
        >>> for move in moves
        """
```

Instruction 2

```python
def _check_five_in_a_row(self, row, col, direction):
        """
        Probe for a continuous line of five same symbols from a specified starting
        cell in a given direction. Inputs 'row' and 'col' designate the start cell
        position, while 'direction', provided as a tuple, specifies the examination
        direction. The process starts by identifying the symbol at the starting cell
        and counting begins. It checks each of the next four cells in the stated
        direction for being within board limits and hosting the identical symbol.
        If a full count of five identical symbols is achieved, it yields 'True', else 'False'.
        :param row: int, row of the given cell
        :param col: int, column of the given cell
        :param direction: tuple, (int, int), named as (dx, dy). Row and col will
                                        plus several dx and dy repectively.
        :return: True if there are five consecutive symbols of the same player,
                                        and False otherwise.
        """
```

Instruction 3

```python
def _check_five_in_a_row(self, row, col, direction):
        """
        Assess whether a sequence of five consistent player symbols can be found starting
        from a chosen cell using a specified direction. The function uses 'row' and 'col'
        to determine the start point and 'direction', a tuple, for the continuity direction.
        It first grabs the symbol at the starting spot and starts counting. For the next four
        cells along the given direction, it checks for board boundaries and identical symbol
        presence. The function returns 'True' for a complete matching line of five, or 'False'
        otherwise.
        :param row: int, row of the given cell
        :param col: int, column of the given cell
        :param direction: tuple, (int, int), named as (dx, dy). Row and col will plus several dx
                                        and dy repectively.
        :return: True if there are five consecutive symbols of the same player, and False otherwise.
        """
```

46. In a few words, what is the task this function aims to implement? *

____

47. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- [ ] Very Easy
- [ ] Easy
- [ ] Not Easy Nor Difficult
- [ ] Difficult
- [ ] Very Difficult

48. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- 1 min
- 5 min
- 10 min
- 20 min
- 40 min
- > 40 min

49. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Demographics Information

50. What is your highest education qualification? *

Une seule réponse possible.

- Ph.D.
- Master's Degree
- Bachelor Degree
- Autre :

51. What is your field of work? (Engineer, Researcher, Developer, etc.) *

52. How many years of development experience do you have? *

Une seule réponse possible.

- Less than 5
- Between 5 and 10
- More than 10

53. Thank you for completing the survey! Any feedback you would like to provide?